

Notes: Moen (JPE, 1997)

Competitive Search Equilibrium

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1 Big picture

- **Question.** Can a market with two-sided search frictions generate a socially efficient unemployment rate and allocation of vacancies without imposing a Hosios bargaining condition exogenously?
- **Core answer.** Yes. Moen introduces the **competitive search equilibrium**, in which workers and firms sort across submarkets indexed by wages and labor-market tightness. In this equilibrium, the private and social returns to entry coincide.
- **Main friction.** Matching is costly and time-consuming. Workers and vacancies meet according to a matching function, so the probability of matching depends on the ratio of vacancies to unemployed workers in the relevant submarket.
- **Main mechanism.** Wages act like prices that decentralize search externalities. A low-wage submarket must compensate workers with high labor-market tightness, while a high-wage submarket can attract workers even with lower tightness.
- **No-surplus logic.** The equilibrium wage in each submarket maximizes the value of vacancies subject to delivering workers a common unemployment value. Otherwise, a new submarket or club could enter and make at least one side better off without making the other side worse off.
- **Optimality result.** Every equilibrium satisfying the competitive-search conditions is socially optimal. This contrasts with ex post bargaining models, where efficiency requires a knife-edge Hosios sharing condition.
- **Implementation result.** Competitive search can be decentralized when firms publicly and credibly announce wages before workers apply. Announced wages segment the market into wage-specific submarkets.
- **Wage dispersion.** Heterogeneous firms offer different wages. More productive firms generally offer higher wages because waiting for a worker is more costly when productivity is high.
- **Worker heterogeneity.** When workers differ in unemployment income or waiting costs, submarkets sort both firms and workers. High unemployment-income workers search in higher-wage, lower-tightness submarkets.
- **Economic takeaway.** Search frictions do not necessarily imply inefficient unemployment. The inefficiency in standard search-and-bargaining models comes from the wage-setting institution, not from matching frictions alone.

2 Incremental contribution of the paper

2.1 What came before

The standard matching framework of Diamond, Mortensen, and Pissarides modeled frictions with a matching function and then usually assumed that wages are determined by ex post bargaining after a match. In that setting, search behavior creates externalities: an extra vacancy helps workers but hurts other vacancies, and an extra worker helps vacancies but hurts other workers. The classic Hosios result shows that the equilibrium is efficient only if workers' bargaining power equals the elasticity of the matching function with respect to unemployment.

2.2 What Moen adds

- **A competitive equilibrium concept for frictional markets.** The paper defines a market equilibrium that preserves the price-taking flavor of Walrasian equilibrium even though agents meet through a matching function.

- **Submarkets as missing markets.** Rather than one labor market with one wage, Moen allows submarkets that differ in wages and tightness. This creates a market for trading off wages against job-finding probabilities.
- **Efficiency without ex post bargaining.** The paper shows that the efficient allocation can be decentralized without choosing a bargaining parameter equal to a matching elasticity.
- **Microfoundation for wage posting.** Public wage announcements implement the competitive search allocation: workers direct their applications toward advertised wages, and firms choose wages anticipating the worker-arrival rate.
- **Sorting and wage dispersion.** The model explains wage dispersion and productivity-wage sorting without appealing to worker misperceptions or purely exogenous efficiency wages.
- **Reinterpretation of efficiency wages.** Firms may pay high wages to attract workers faster, but the induced unemployment can still be efficient. High wages are not automatically evidence of market failure.

2.3 Why the contribution matters

The paper changes the interpretation of search externalities. Matching frictions are real, but if firms can post wages and workers can direct search, the price system can internalize the externalities through the wage-tightness trade-off. This insight became central to the later literature on directed search, wage posting, and competitive search.

3 Model primitives and measurement

3.1 Agents, timing, and matching technology

- Time is continuous.
- The measure of workers is normalized to one.
- Firms can open vacancies by paying sunk cost k .
- After paying the vacancy creation cost, the vacancy draws productivity y_i from a discrete distribution with probabilities f_i .
- The matching function $x(u, v)$ maps unemployed workers u and vacancies v into new matches.
- The matching function is concave, constant returns to scale, and differentiable.

Define labor-market tightness as

$$\theta = \frac{v}{u}.$$

The worker job-finding rate and vacancy worker-arrival rate are

$$p(\theta) = x(1, \theta), \quad q(\theta) = x(1/\theta, 1).$$

Interpretation: Higher θ means more vacancies per worker. Workers find jobs faster, but firms wait longer for workers. The entire competitive-search mechanism is built on this trade-off.

3.2 Submarkets

A submarket consists of a subset of unemployed workers and vacancies that search for each other. In the market-maker formulation, each submarket has one wage w_i and one tightness θ_i .

- Workers are free to choose the submarket that maximizes expected income.

- Firms are free to choose the submarket that maximizes vacancy value.
- Since workers are identical in the baseline, all active submarkets must give workers the same expected unemployment value U .

Interpretation: Submarkets are the key institutional device. They turn an undirected search problem into a directed search problem, where firms and workers choose among wage-tightness packages.

3.3 Worker value equations

Let U_i be the asset value of unemployment in submarket i and E_i the value of employment at wage w_i . Then

$$rU_i = z + p(\theta_i)(E_i - U_i), \quad (1)$$

$$rE_i = w_i - s(E_i - U_i). \quad (2)$$

Substituting out E_i gives

$$rU_i = \frac{(r+s)z + w_i p(\theta_i)}{r+s+p(\theta_i)}.$$

Because all active submarkets deliver the same U , the worker indifference condition is

$$p(\theta_i) = \frac{rU - z}{w_i - rU}(r+s).$$

This defines the worker indifference curve

$$\theta_i = \theta(w_i; U).$$

Interpretation: Workers accept a low wage only if the job-finding rate is high. Thus a low wage must be paired with high tightness from the worker perspective.

3.4 Firm value equations

For a vacancy of productivity y_i , let $V(y_i, w, \theta)$ be the vacancy value and $J(y_i, w)$ the value of a filled job. Then

$$rV(y_i, w, \theta) = -c + q(\theta)[J(y_i, w) - V(y_i, w, \theta)], \quad (3)$$

$$rJ(y_i, w) = y_i - w - sJ(y_i, w). \quad (4)$$

Substituting out J yields

$$(r + q(\theta))V(y_i, w, \theta) = q(\theta)\frac{y_i - w}{r + s} - c.$$

Interpretation: Firms dislike high wages and high tightness. High wages reduce output net of labor cost; high tightness means many vacancies per worker, so the arrival rate $q(\theta)$ is low.

4 Equilibrium and optimality conditions

4.1 No-surplus wage choice

A wage w in an active submarket must solve

$$w_i \in \arg \max_w V(y_i, w, \theta(w; U)).$$

This is the no-surplus condition: no alternative wage-tightness package can attract workers while making a vacancy type strictly better off.

Interpretation: The wage is not set by bargaining after a match. It is set ex ante as the price that trades off wage cost against the probability of filling the vacancy.

4.2 Entry and steady-state equations

Let $\hat{i}$ be the lowest productivity type that opens a vacancy. The expected value of vacancy creation is

$$\bar{V}(U) = \sum_{i=\hat{i}}^n f_i \max_w V(y_i, w, \theta(w; U)).$$

Free entry requires

$$\bar{V}(U) = k.$$

The equilibrium system is

$$\bar{V}(U) = k, \tag{5}$$

$$w_i = \arg \max_w V(y_i, w, \theta(w; U)), \quad i \geq \hat{i}, \tag{6}$$

$$rU = \frac{(r+s)z + p(\theta_i)w_i}{r+s+p(\theta_i)}, \quad i \geq \hat{i}, \tag{7}$$

$$u_i p(\theta_i) = \tilde{f}_i (1-u)s, \quad i \geq \hat{i}, \tag{8}$$

$$\sum_{i=\hat{i}}^n u_i = u. \tag{9}$$

Interpretation: The model is almost recursive. Entry determines U ; U determines wage-tightness pairs; steady-state flows determine unemployment and the distribution of searchers.

4.3 Existence

Proposition 1 states that an equilibrium exists if the economy is productive enough:

$$\sum_{i=1}^n f_i \frac{\max[y_i - z, 0]}{r+s} > k.$$

Interpretation: If expected productivity net of unemployment value is too small, vacancies are not worth opening. The condition guarantees that positive entry can occur.

4.4 Social planner problem

The social planner maximizes discounted aggregate production net of unemployment income, vacancy search costs, and vacancy creation costs:

$$W = \int_0^\infty e^{-rt} \sum_{i=\hat{i}}^n (N_i y_i + z u_i - c v_i - a k) dt,$$

subject to transition equations for employed workers and vacancies:

$$\dot{N}_i = v_i q(v_i/u_i) - s(1 - N_i), \tag{10}$$

$$\dot{v}_i = a f_i - v_i q(v_i/u_i), \tag{11}$$

$$\sum_{i=\hat{i}}^n (u_i + N_i) = 1. \tag{12}$$

Proposition 2 states that all equilibria satisfying the competitive-search system are optimal.

Interpretation: Efficiency has two parts: the allocation of workers and vacancies across submarkets is efficient, and the number of vacancies opened is efficient.

4.5 Connection to the Hosios condition

Let the match surplus be

$$S_i = E_i + J_i - U - V_i,$$

and let

$$\eta_i = \frac{\theta_i q'(\theta_i)}{q(\theta_i)}.$$

The equilibrium can be rewritten as

$$rV_i = -c + q(\theta_i)(1 - \eta_i)S_i, \quad (13)$$

$$S_i = \frac{y_i + sU}{r + s} - U - V_i. \quad (14)$$

Interpretation: These equations resemble the search-and-bargaining model when workers' bargaining power equals the matching elasticity. Moen's point is that competitive search produces this internalization through wage posting rather than assuming the right bargaining share.

5 Wage announcements and implementation

5.1 Wage posting as submarket formation

If firms publicly announce wages before workers apply, each announced wage forms a submarket. Workers choose which wage category to apply to. In equilibrium, all active wage categories deliver the same U to workers.

Interpretation: Wage advertisements decentralize the market-maker allocation. They communicate the wage and let workers direct their search.

5.2 Firm beliefs and stability

Firms choose wages based on beliefs about how the announced wage affects worker arrival rates. A rational-expectations equilibrium must satisfy the correct beliefs on the equilibrium path. Moen argues that stability requires the same wage-tightness relationship off the equilibrium path:

$$q^e(w) = q(\theta(w; U)).$$

Interpretation: If a deviating wage would actually attract workers according to $\theta(w; U)$, then beliefs inconsistent with this relationship are unstable. A small group of deviating firms would profit and change the equilibrium.

5.3 Equivalence result

Proposition 3 states that, under the described beliefs, the set of announced-wage equilibria and the set of competitive search equilibria are equivalent.

Interpretation: The abstract market-maker equilibrium is not merely a planner construct. It can be implemented by wage-posting behavior under plausible assumptions about worker-directed search.

6 Extensions and comparative implications

6.1 Productivity and wages

Let the equilibrium wage of a vacancy of productivity y be denoted

$$w = \delta(y; U).$$

Proposition 4 states that $\delta(y)$ is strictly increasing in productivity. If $c = 0$, the limiting wage as productivity approaches the minimum active level is rU .

Interpretation: High-productivity firms have a higher opportunity cost of waiting, so they pay higher wages to attract workers faster. The model therefore generates wage dispersion from firm productivity differences.

6.2 Heterogeneous workers

When workers differ in unemployment income z_i , each type has its own indifference curve $\theta_i(w; U_i)$. The equilibrium may contain submarkets sorted by both firm productivity and worker outside options.

Interpretation: Worker types with high unemployment income suffer less from waiting, so they are more willing to enter higher-wage, lower-tightness submarkets. Workers with low unemployment income value quick employment more and enter lower-wage, higher-tightness submarkets.

6.3 Empirical implication

The model predicts that firms offering higher wages should attract more applicants and fill vacancies faster, holding job characteristics constant.

Interpretation: The empirical challenge is that high-wage jobs often also require rarer skills. Still, the mechanism points to a direct link between wage offers and applicant queues.

7 Figures, proposition blocks, and interpretation

The paper has no empirical tables. The important visuals are theoretical diagrams. Adjacent figures and theoretical blocks are arranged compactly in pairs where possible.

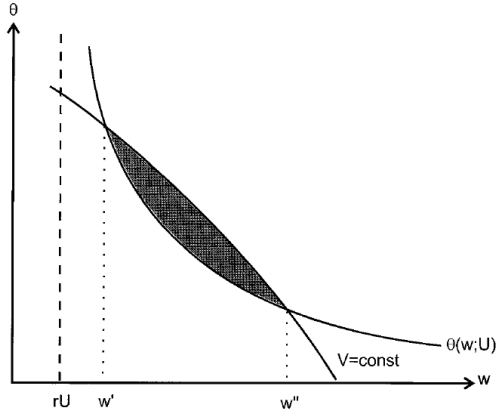
7.1 Figure 1: Non-tangency and the no-surplus failure; Figure 2: Tangency equilibrium

Read:

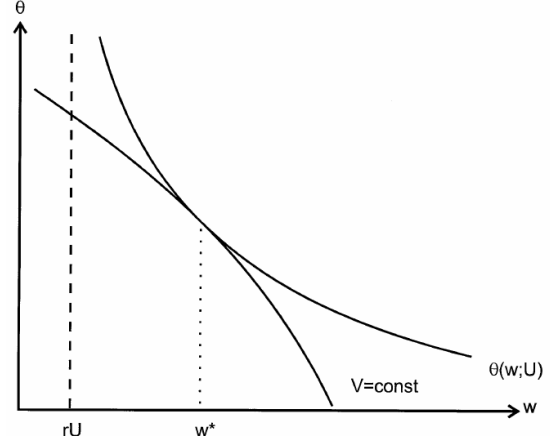
- Figure 1 shows a candidate wage w' where the worker indifference curve and the firm iso-profit curve do not touch tangentially.
- The shaded area contains wage-tightness combinations that improve both sides: workers get the same or higher value, and firms get higher vacancy value.

- Such a point cannot be a no-surplus equilibrium because an entering submarket could exploit the shaded area.
- Figure 2 shows the equilibrium point w^* , where the curves are tangent.
- At tangency, the marginal rate of substitution between wages and tightness is equal for workers and firms.

Economic message: The no-surplus condition transforms search equilibrium into a price system. The equilibrium wage is the price that makes workers' valuation of tightness and firms' valuation of tightness coincide.



(a) Figure 1: no-surplus failure when curves are not tangent.



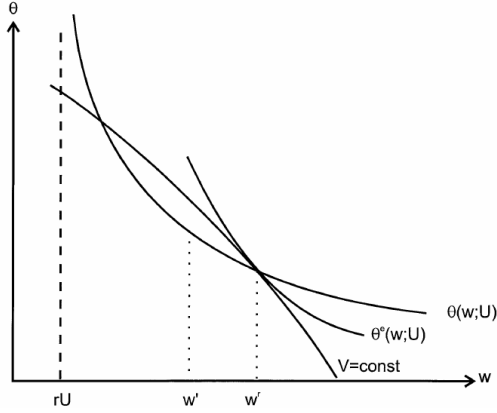
(b) Figure 2: competitive search equilibrium at tangency.

7.2 Figure 3: Unstable wage-announcement beliefs; Figure 4: Wage sorting with heterogeneous firms

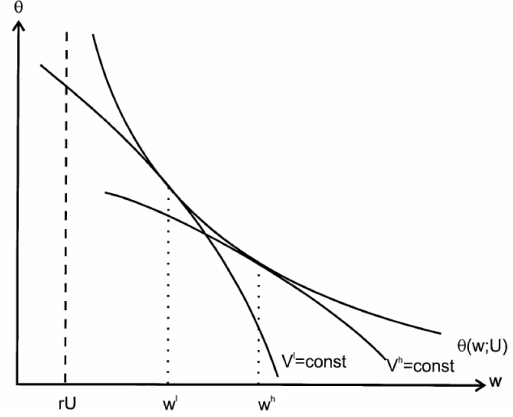
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- Figure 3 compares arbitrary out-of-equilibrium beliefs $\theta^e(w; U)$ with the true worker indifference relationship $\theta(w; U)$.
- If firms announce w' because of incorrect beliefs, a small deviation to w^* can yield higher vacancy value.
- The instability supports the equilibrium refinement requiring firms' beliefs to match the worker indifference curve off path.
- Figure 4 shows two productivity types. The high-productivity firm's iso-profit curve is flatter because waiting is more costly for high productivity firms.
- The high-productivity firm offers a higher wage and operates in a lower-tightness submarket than the low-productivity firm.

Economic message: Wage posting implements competitive search only if workers can direct applications and firms understand the wage-applicant trade-off. Heterogeneous firm productivity then maps into wage dispersion.



(a) Figure 3: wage-announcement equilibrium with unstable beliefs.



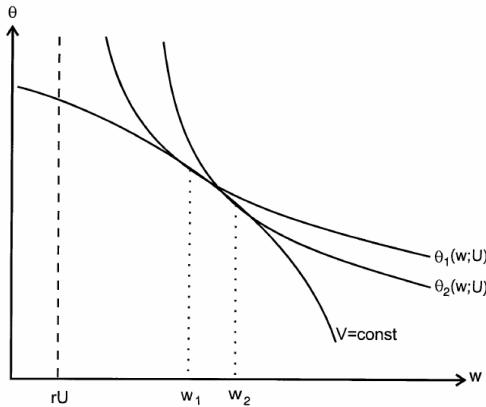
(b) Figure 4: heterogeneous firms and wage sorting.

7.3 Figure 5: Heterogeneous workers with homogeneous firms; Figure 6: Heterogeneous workers and firms

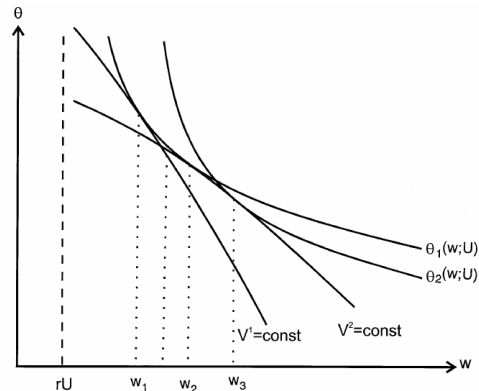
Read:

- Figure 5 shows two worker types with different unemployment income. Each type has its own indifference curve.
- The equilibrium separates worker types into different submarkets even though firms are homogeneous.
- Low-unemployment-income workers value quick matching more, so they accept lower wages in markets with higher tightness.
- Figure 6 adds firm heterogeneity. The model generates three active submarkets because firm productivity and worker outside options jointly determine the efficient pairing.
- High-productivity firms and high-unemployment-income workers can meet in higher-wage, lower-tightness submarkets; low-productivity firms and low-unemployment-income workers sort to lower-wage, higher-tightness submarkets.

Economic message: Competitive search is a sorting theory. Wages do not just divide surplus after matching; they direct workers and firms into the right queues before matching.



(a) Figure 5: heterogeneous workers, homogeneous firms.



(b) Figure 6: heterogeneous workers and heterogeneous firms.

7.4 Key propositions and equation blocks: equilibrium system, optimality, implementation, and sorting

Read:

- **Equations (10)–(14)** collect the competitive-search equilibrium system: free entry, wage maximization, worker indifference, flow balance, and unemployment accounting.
- **Equations (19)–(20)** rewrite the model in a Hosios-like form, showing how the competitive-search allocation resembles a bargaining model with the efficient bargaining share.
- **Proposition 1** gives the productivity condition for existence: the economy must be productive enough for vacancy creation to be worthwhile.
- **Proposition 2** gives the central welfare theorem: all equilibria satisfying the competitive-search system are optimal.
- **Proposition 3** says that, under stable wage-announcement beliefs, announced-wage equilibria and competitive search equilibria are equivalent.
- **Proposition 4** shows that the equilibrium wage schedule is increasing in firm productivity.
- **Proposition 5** summarizes sorting with heterogeneity: high-productivity firms and high-unemployment-income workers sort into higher-wage submarkets.

Economic message: The proposition sequence is the paper’s logic in compressed form. Proposition 1 establishes existence, Proposition 2 establishes efficiency, Proposition 3 gives decentralization through wage announcements, and Propositions 4–5 show that the model generates wage dispersion and sorting once firms and workers differ.

Clean proposition map.

- **Existence:** productive opportunities must cover vacancy creation costs.
- **Optimality:** the asset value of a vacancy equals its social value; therefore entry and the allocation across submarkets are efficient.
- **Implementation:** posted wages generate submarkets; stable beliefs about off-path wages select the competitive-search allocation.
- **Sorting:** high-productivity firms post higher wages; workers with higher outside options search in higher-wage submarkets.

8 Short summary

- Moen introduces competitive search equilibrium as a price-taking equilibrium concept for markets with matching frictions.
- Submarkets are indexed by wages and labor-market tightness.
- Workers choose submarkets that maximize expected income; firms choose submarkets that maximize vacancy value.
- Worker indifference implies a downward-sloping relationship between wages and tightness.
- Firms face a trade-off between paying higher wages and filling vacancies faster.
- The no-surplus condition requires wages to maximize vacancy value subject to worker expected income.
- Free entry pins down the common unemployment value U .
- The competitive search allocation is socially efficient.
- Wage announcements can implement the competitive search equilibrium if firms publicly post wages and workers direct applications.

- High-productivity firms post higher wages because waiting is more costly for them.
- Worker heterogeneity leads to sorting across submarkets by outside options and waiting costs.
- The paper’s main lesson is that search frictions do not automatically imply inefficiency; the institution of wage setting is crucial.

9 Conclusion

9.1 Core conclusion

Moen’s paper establishes that a frictional labor market can behave like a competitive market when wages are posted before search and workers can direct their applications. The competitive search equilibrium is efficient because wages and queue lengths jointly price the externalities created by search. A vacancy’s private value coincides with its social value because the submarket wage forces the firm to internalize the worker’s required expected income.

9.2 Economic intuition

The model turns unemployment and queues into prices. A low wage is not just a low transfer to workers; it also attracts fewer workers relative to vacancies from the worker perspective, creating a faster job-finding probability. A high wage attracts more workers and allows a firm to fill the job faster. This wage-tightness trade-off plays the role of a market-clearing price for search time. When the curves are tangent, no new submarket can improve on the allocation.

9.3 Incremental contribution in one sentence

The paper’s incremental contribution is to replace ex post bargaining in matching models with ex ante wage posting and directed search, thereby generating efficient unemployment and wage dispersion through a competitive equilibrium mechanism rather than through an imposed bargaining-share condition.

9.4 Relation to Hosios and search-and-bargaining models

The Hosios condition says that the worker’s bargaining share must equal a matching-function elasticity for efficiency. Moen shows a different route: wage posting makes firms choose wages that directly balance wage costs against worker-arrival rates. This decentralizes the efficient allocation without requiring bargaining power to take a particular value. In homogeneous-firm models, the competitive-search equations can be rewritten to look like the Hosios-efficient bargaining equations; with heterogeneous firms, competitive search does more because it separates different productivity types into different submarkets.

9.5 Why the implementation result matters

The market-maker formulation is useful for theory but artificial. The wage-announcement section shows how ordinary labor-market institutions can implement the same allocation. If firms advertise wages and workers can choose which wage category to apply to, then wage categories become submarkets. This gives the model empirical content: high wages should attract more applicants and reduce vacancy duration, all else equal.

9.6 Limitations and open questions

The model abstracts from many features of real labor markets: match-specific productivity, multidimensional jobs, on-the-job search, private information about worker quality, wage renegotiation, finite worker application

capacity, and endogenous information acquisition. It also assumes that posted wages are credible and that workers observe wages early enough to direct search. These assumptions are plausible in some labor markets but less plausible in markets where compensation is negotiated late or where vacancies bundle many unobserved nonwage characteristics.

9.7 Takeaway

The paper's lasting insight is that market frictions are not synonymous with market failure. If prices are posted before matching and agents can direct search, wages can coordinate both sides of the market and internalize search externalities. This is why competitive search became a central framework for studying labor markets, housing markets, retail markets, and other environments where agents choose not just prices but also queues.